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Bachelor of Computer Applications

Semester – V

Business Analytics

Experiment No.: 4

Title: Data Cleaning using Power Query

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Data Cleaning using Power Query

GitHub Repository Link: https://github.com/nidhidev2122/Business_analytics

Problem Statement

A company has received customer data containing missing values, duplicate records, and inconsistent entries. Clean and prepare the dataset using Power Query before analysis.

Dataset used: Cafe Sales Dirty Data

Source: <https://www.kaggle.com/datasets/ahmedmohamed2003/cafe-sales-dirty-data-for-cleaning-training>

Steps Followed:

Step 1: Data Collection

The Cafe Sales Dirty Data dataset was used for this experiment. It was obtained from Kaggle and downloaded in CSV format. The dataset contains cafe sales transactions with details such as:

- Transaction ID, Item, and Quantity
- Price per Unit and Total Spent
- Payment Method and Location
- Transaction Date

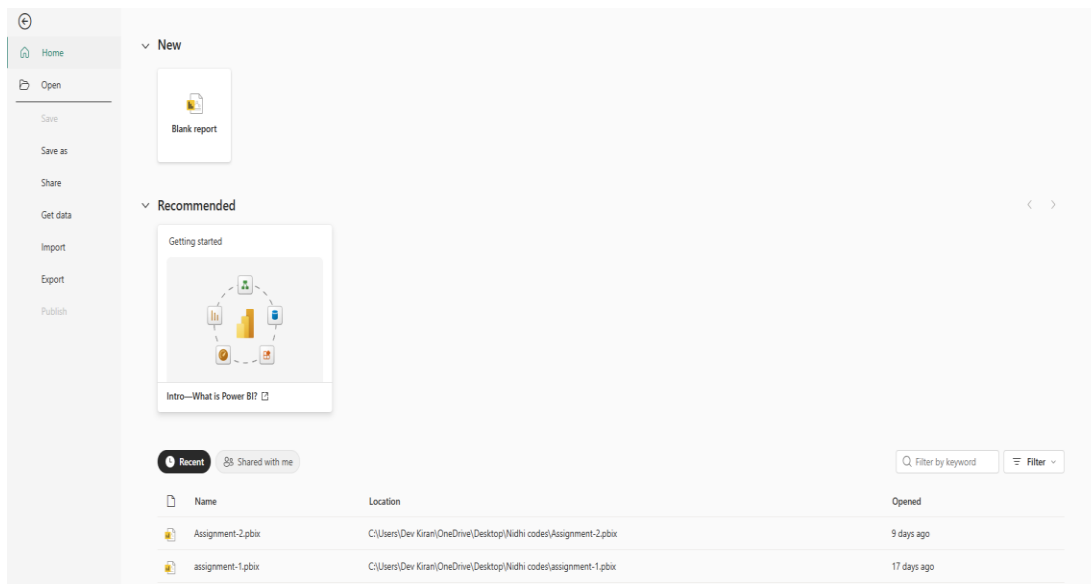


Figure 1: Power BI Desktop Home Screen

Step 2: Data Import

The file `dirty_cafe_sales.csv` was loaded directly as a flat file. Power Query Editor was launched to inspect and clean the data before analysis.

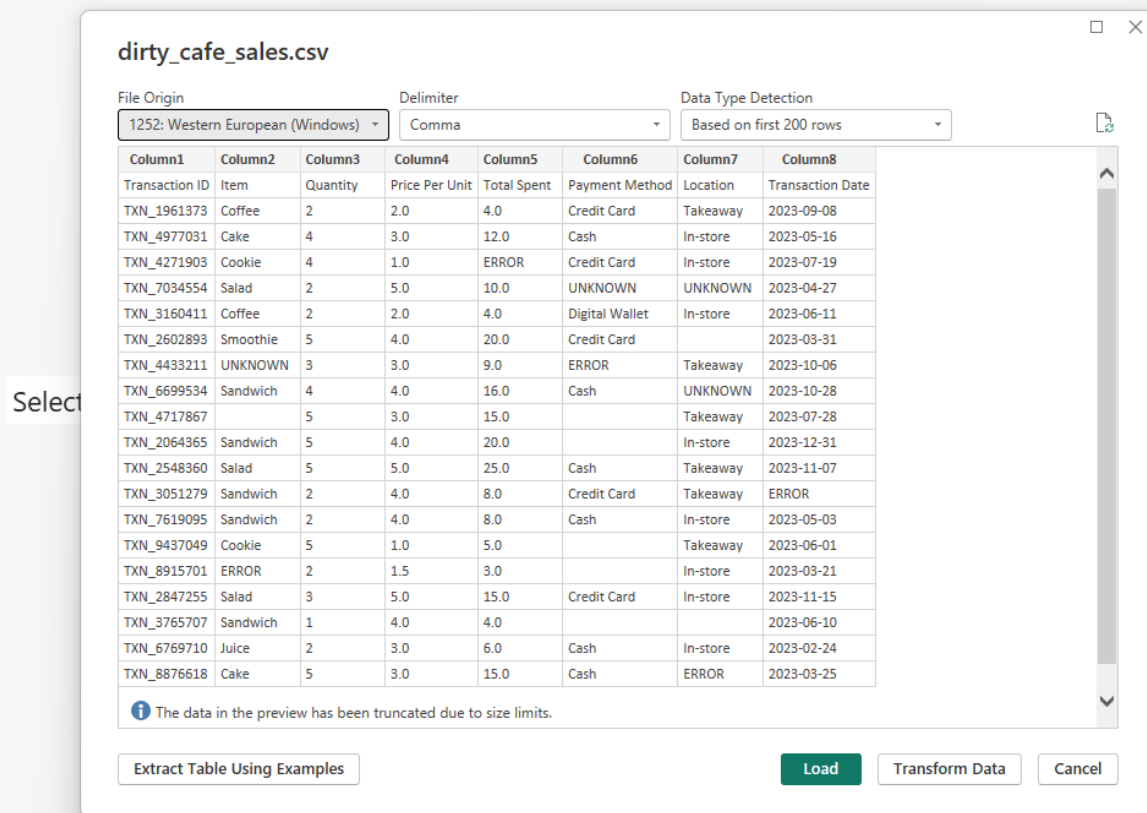


Figure 2: Get Data

Step 3: Data Cleaning

3.1 Header Promotion

The first row of the dataset was set as the column headers using the "Use First Row as Headers" function.

3.2 Correcting Data Types

Several columns had incorrect data types (mostly stored as text) and were converted appropriately:

- Quantity: converted from Text to Whole Number
- Price Per Unit: converted from Text to Fixed Decimal Number
- Total Spent: converted from Text to Fixed Decimal Number

- Transaction Date: converted from Text to Date format

3.3 Removing Invalid Entries

The Item column contained several invalid or non-informative values, which were removed:

- Entries marked as "UNKNOWN"
- Entries marked as "ERROR"
- Empty/blank entries

3.4 Eliminating Incomplete Rows

Rows containing missing or null values in the following key columns were dropped, since these fields were essential for analysis:

- Quantity
- Price Per Unit
- Total Spent
- Transaction Date

3.5 Fixing Inconsistent Categorical Data

The Payment Method and Location columns had inconsistent placeholder values (such as "ERROR", "UNKNOWN", and blanks). Rather than deleting these rows, the inconsistent values were standardized by replacing them with "UNKNOWN" — this preserved the transaction records while keeping the categorical data clean and uniform.

3.6 Checking for Duplicates

The TransactionID column was scanned for duplicate entries using the "Remove Duplicates" feature. No duplicates were detected — all 10,000 Transaction IDs were confirmed unique.

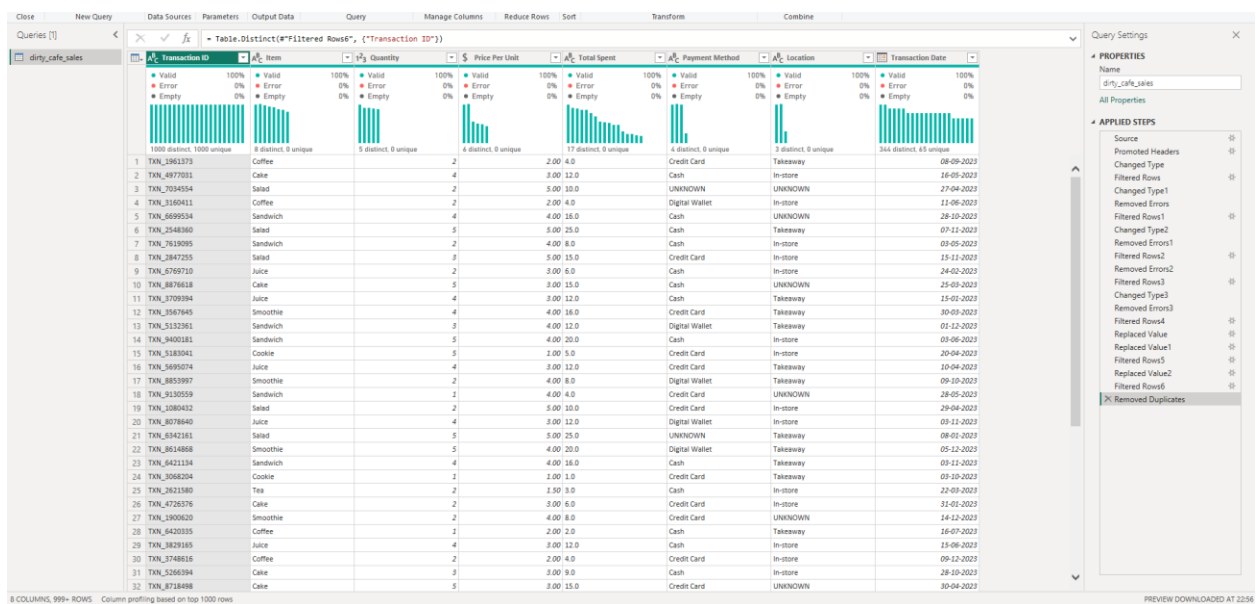


Figure 3: Power Query Editor - Applied Steps + Cleaned Data

Summary:

- **Transaction ID:** All transaction IDs were valid, with no missing or erroneous entries.
- **Item:** Invalid entries such as UNKNOWN, ERROR, and blank values were removed. The final dataset contains 8 valid items: Coffee, Tea, Juice, Cake, Cookie, Sandwich, Salad, and Smoothie.
- **Quantity:** Invalid and missing values were removed, and the column was converted to **Whole Number** format.
- **Price Per Unit:** Incorrect and missing entries were removed, and the values were converted to **Fixed Decimal Number** format.
- **Total Spent:** Error, blank, and invalid values were removed, followed by conversion to **Fixed Decimal Number**.
- **Payment Method:** Missing, blank, and error entries were standardized as UNKNOWN, while valid payment types such as Cash, Credit Card, and Digital Wallet were retained.
- **Location:** Inconsistent or missing location values were replaced with UNKNOWN. Valid values such as In-store and Takeaway were retained.
- **Transaction Date:** Invalid and missing dates were removed, and the column was converted to the **Date** data type.

After completing the cleaning process, **all 8 columns contained 100% valid values with no errors or empty entries**, as confirmed using the Column Quality feature in Power Query.

Conclusion:

This experiment provided practical experience in using **Power Query in Power BI** to prepare an unclean dataset for analysis. The Cafe Sales dataset initially contained **10,000 records** with several types of data quality problems. These issues were addressed by promoting the headers, correcting column data types, removing invalid records, handling missing values, standardizing inconsistent information, and checking for duplicate transactions.

After the cleaning process, the dataset contained **7,382 usable records**, making it more suitable for further analysis. The experiment demonstrated how proper data preparation can improve the reliability and accuracy of the results obtained from business data.

Business Insights:

The following key business insights were derived from the cleaned Cafe Sales dataset:

- **Overall Data Quality:** Out of the original 10,000 records, **2,618 records contained issues**, showing that a considerable portion of the raw dataset required cleaning before analysis.
- **Product Information:** The cleaned data contains **8 cafe products**, allowing sales performance to be compared across different items.
- **Payment Methods:** Missing and inconsistent payment information was observed, highlighting the importance of maintaining accurate transaction records.
- **Location Information:** Some location values were incomplete or invalid, indicating that transaction location details need to be recorded more consistently.
- **Data Preparedness:** The final **7,382 records** were standardized across all 8 columns, making the dataset suitable for creating reports and performing business analysis.
- **Transaction Integrity:** The Transaction ID check showed **no duplicate records**, confirming that each transaction was uniquely identified.